Calibration guide

The monitor, the probe, the ICC profile: the steps, in the order you do them.



FIG. 1 The probe on the white measurement square while the application runs the plan. Original render, not a product photo.

Cristi Gordaș · free edition, GPLv3
Based on DisplayCAL (Florian Höch, community continuation) and ArgyllCMS (Graeme Gill).

DisplayCAL-CG User Guide

Monitor calibration and profiling, step by step — GDC edition

Document version 2.0 · based on DisplayCAL-CG 3.10.0.dev82-cg.4

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1. What is DisplayCAL-CG

DisplayCAL-CG is a GDC edition of the open-source DisplayCAL project (the community continuation of Florian Höch's original work), repackaged and fully translated by GDC. The application calibrates and profiles your monitor using the ArgyllCMS measurement engine — the result is a monitor with correct, consistent colors no matter which application you're working in (photo, video, graphics).

> **Calibration vs. profiling — what's the difference**
> Calibration brings the monitor to a known target state (white point, luminance, tone curve) through hardware/software adjustments (the video card's curves). Profiling then measures how the calibrated monitor responds and writes an ICC profile that the rest of the system uses for accurate color correction. They're always done one after the other, in this order.

This guide walks through each of the 5 main tabs (Display & instrument, Calibration, Profiling, 3D LUT, Verification), the advanced tools available from the menus, and how to install/update the application on Mac and Windows.

> **You need a measurement device**
> DisplayCAL-CG CANNOT calibrate a monitor without a physical colorimeter or spectrophotometer connected via USB (e.g. X-Rite i1Display Pro, ColorMunki Display, Datacolor Spyder). The app auto-detects the connected device on the "Display & instrument" tab.

2. Installation

macOS

1. Download the DisplayCAL-CG.pkg package from the download page (gordas.dev/DisplayCAL-CG).
2. Double-click the downloaded .pkg file.
3. In the installer window, click Continue on each step until you reach the License page.
4. Read the GPLv3 license summary shown, click Agree (without explicit acceptance, the install cannot proceed — this is a requirement of the native macOS installer, not an optional step).
5. Click Install — the app and all 9 satellite tools (Profile Info, Curve Viewer, etc.) are installed directly into Applications, nothing to drag manually.
6. Open DisplayCAL-CG from Launchpad/Applications.

> **The package is Apple-signed and notarized**
> macOS Gatekeeper will let the app launch directly, without the "cannot be opened because it is from an unidentified developer" warning.

Windows

1. Download the DisplayCAL-CG-Setup.exe installer from the download page.
2. Double-click the downloaded file.
3. If the Windows SmartScreen warning appears ("Windows protected your PC"), click More info, then Run anyway.
4. On the installer's license page, select I accept the agreement — the "Next" button stays disabled until you check this.
5. Follow the installer's steps (the default location is recommended) through to Install.
6. Open DisplayCAL-CG from the Start Menu or the Desktop shortcut.

Why the SmartScreen warning appears
The Windows installer isn't (yet) signed with a paid code-signing certificate — the warning is normal for unsigned software, not a sign that the file is unsafe. Only download it from the official page above.

3. Guided calibration (5 steps)

The first tab, Guided calibration, takes you from a connected instrument to a verified profile in five steps. All advanced options stay available in the other tabs.

1. Monitor and instrument — the application shows what it detected. Some instruments share one USB identity (for example the i1 DisplayPro and ColorMunki Display), so it cannot tell the exact model by itself: you confirm it, and every line says whether it was detected by the system or confirmed by you.
2. Your goal — pick the purpose (photo, video, general use…). Each goal maps to a ready-made preset.
3. Preparation — a short checklist for the things the software cannot do for you (see below). Nothing is ticked automatically.
4. Measurement — a summary of what will happen, the measurement plan, and the start button.
5. Install and verify — opens only after a valid profile exists. After installing, the application reads back the active profile and tells you whether it is really the one that was just created.

Text size
The Aa button (also in the Options menu) sets the text size: Normal, Large or Very large. It starts from the size your system uses, applies at once to the whole application including explanations and dialogs, and can be reset to the default. It changes only the text of the application; the measurement window, the patch size and colours are not affected.

If something is missing
When ArgyllCMS, a display or an instrument is not found, the first step says so plainly and what to do. The Argyll engine is included in the installer (verified against a fixed SHA-256 hash); if a download is ever needed, a file that does not match the expected hash is refused.

4. Physical preparation

Before measuring, the checklist covers the monitor (stable, final settings), the room (no light falling on the screen), and the instrument (correct placement, cable, cap). Each item has a short why with its sources, and you tick them yourself.

No universal warm-up time
There is no single warm-up time valid for every monitor. The application does not invent one: it shows a figure only when the instrument maker publishes it (for example 30 minutes for Datacolor SpyderX), and presents it as that maker's recommendation.

For the instruments whose placement is documented by the maker (Calibrite ColorChecker Display, Pro, Plus and Datacolor SpyderX) a short animation shows the steps, with a text version of each step; play/pause and previous/next buttons control it, and it respects the system “reduce motion” setting. For other instruments the steps are generic and say so. The animation always stops before measurement starts; the measurement window itself shows only neutral patches.

5. Measurement plan and long sessions

Step 4 shows the plan before anything starts, with three numbers kept apart: the total patches in the chart, the largest block and the number of blocks (for example “11,000 patches in total · blocks of at most 500 · about 22 blocks”), plus the estimated time. The same plan appears in the Profiling tab, and the button Choose or create a chart beside the total opens the chart file chooser or the chart editor. The block size never changes the total: it only decides how the same chart is cut into pieces.

- Profile type — a matrix profile (one shared curve or per-channel curves) is small and predictable; a LUT profile (XYZ or Lab) describes more of the screen's behaviour but needs more patches. The profile type, the calibration curves and the measurement chart are three different things: curves adjust the monitor, the chart is what is measured, the type is how the result is described to applications. Macs: on macOS 10.8 or newer the application asks once if the type is anything other than single curve + matrix with black point compensation, because, according to the original project, applications that use Apple's own colour module (Preview, Quick Look) may show black crush and posterization with other display profile types. It is a question, not a ban: Yes switches, No keeps your choice. Applications with their own colour management (Adobe, GIMP, Krita) and workflows using a 3D LUT are not affected. More patches is not automatically better: it costs time, lets the display drift and only helps where the profile type can use it; verify the result afterwards.
- Calibration vs profiling — calibration adjusts the screen towards a target; profiling measures it and writes the ICC profile. The chart (the set of patches) belongs to profiling.
- Measurement in blocks with recovery — a large chart can be measured in blocks of at most 100, 250, 500, 1000 or 2000 patches (or another size you type). There is no fixed upper limit on the chart itself: charts of 11,000 patches or more can be measured this way. A finished block is saved

to disk; if the run is interrupted, the finished blocks are kept and you can try to continue (recovery is not guaranteed in every situation, e.g. a power cut). Each block repeats the white reference, so the run needs one extra white reading per block after the first (shown separately) and re-initialises the instrument between blocks. The time estimate uses the speed this instrument really had in earlier runs on this computer when there is such a history, otherwise its specification, and says which; it is never a guarantee. Instruments that need their own calibration cycle measure in one run.
- Keep the computer awake — on by default, only for the duration of the run, released at the end. It cannot prevent: closing the lid, choosing Sleep, the power button, a flat battery, locking the screen or switching user. Keep the computer plugged in and the session unlocked.

At the end of a long run the application shows a local notification (nothing leaves the computer).

6. If a run stops unexpectedly

While measuring, the application keeps a record on disk (a journal updated continuously, saved atomically, in the application's data folder). It records facts, not guesses: when activity was last seen, progress, which stage, how the measurement engine ended, memory and free disk space, and whether the computer was suspended.

At the next start, if a session never recorded an end, a window lists what is known and offers:

- Export everything recoverable to a folder of your choice.
- Show folder to inspect the files.
- Continue with the remaining blocks — offered only for runs measured in blocks, and only when the monitor, instrument, mode, correction, engine version and profile type are the same. Measurements taken under different conditions are never combined silently. The block that was in progress is measured again from its start.
- Dismiss or mark as seen.

> **What cannot be recovered**
> A run measured as a single block keeps its readings only inside the measurement engine until it ends; if it is interrupted, they are lost. That is why long runs should use blocks.

7. What the results mean, and their limits

A run is only a success when the profile exists, is installed for this monitor and passes verification. Finishing a process proves none of that.

Verification measures a set of patches and compares them with what the profile predicts. It describes those patches, on that monitor, with that instrument, at that moment; it is not a certificate for every colour.

What is what

- Calibration — Adjusts how the monitor itself behaves (brightness, white point, tone curve), through its controls and the graphics-card curves.
- ICC profile — Describes the colours the calibrated monitor shows. Colour-managed applications use it; it does not change the screen by itself.

- Loading the curves — The calibration curves (vcgt) must be loaded into the graphics card at login for the calibration to apply system-wide.
- 3D LUT — An optional correction for video players and grading tools that cannot use ICC profiles. Created separately, from a finished profile.

Limits to keep in mind

- Applications without colour management ignore the ICC profile; only the calibration curves reach them.
- HDR and SDR are different modes: a calibration made in one does not apply to the other, and an operating system in HDR mode may bypass the curves.
- With several monitors, each needs its own run, and the profile is tied to that monitor and that instrument.

Two more limits: a result obtained with one instrument says nothing about another, and a profile made with a long, drifting run is only as good as the stability of the screen during it. Verify after profiling and repeat if the screen has changed.

8. Diagnostic and support

Help → Diagnostic and support gathers what you need when something goes wrong, on macOS and Windows. Everything stays on your computer; the application uploads nothing and asks for no account.

- Log folder — a button opens it. On macOS it is ~/Library/Logs/DisplayCAL; on Windows %LOCALAPPDATA%\DisplayCAL\Logs. The logs rotate daily and when a file reaches 5 MB, and the total is capped, so they cannot fill the disk.
- Versions — application, system, Qt, ArgyllCMS and other components, in a box you can copy.
- Session ID — each run of the application has a short ID that appears on every log line; a measurement session also appears under its own ID. Copy it into your report.
- Detailed logging — can be switched on for the current run only; it switches itself off after 60 minutes and is never kept for the next start.

1. Press Export diagnostic package….
2. A window shows what will be included, with sizes. Always included: versions, the application log, the journals of the last three measurement sessions (stages, how each ArgyllCMS process started and ended, progress, memory, free disk space) and crash traces written by the application. Only if you tick them: measurement files, the display's ICC profile and, on macOS, the system's crash reports about the application. Memory dumps and your personal files are never included.
3. Personal data is removed from the text by default (user name, home folders, e-mail, tokens, network and hardware identifiers). Unticking this asks for confirmation.
4. Choose where to save the ZIP. You can open it with any tool and read it; manifest.json says what is inside and what was left out.

> **Sending it to support is your decision**
> The application never sends the package. If you want help: open the support page (button in the same window), describe the problem and attach the ZIP to the report. If you prefer, look inside first and send only part of it.

After an unexpected stop, the next start notes in the log that the previous run did not shut down normally, and the recovery window (previous chapter) lists the unfinished sessions; both are covered by the package.

9. Advanced settings, grouped by decision

The advanced tabs keep every professional control. Each tab is now divided into sections named after the decision you take (for example What the screen should look like, Type of profile, File format and precision). Under each section title, What each setting here changes opens a list that says, for every setting: what it changes, when it fits, its unit and the application's default, what it depends on, and whether it is required to run, depends on your case or optional.

When a control is unavailable or hidden, the same list says why (for example: a spectrometer does not use a correction; the 3D LUT tab is switched on in the Options menu; the PCS-to-device mode needs a LUT-type profile). The explanation is also in the control's tooltip.

- The guided and the advanced mode work on the same configuration. Opening the advanced tabs and coming back changes nothing. In the guided flow the goal Keep my current settings loads no preset at all; if you changed settings after choosing a preset, the guided flow tells you how many settings differ and offers to load the preset again or to treat them as your own.
- Defaults are shown as the application's default, never as a universal recommendation: the text says in which case a choice fits.
- In the test chart editor every field has a tooltip, taken from the ArgyllCMS targen documentation.

10. Finding your monitor's technology

Next to the colorimeter correction (in the guided flow and in the advanced Display tab) the link Don't know your monitor's technology? Search the model opens a window with the detected model, which you can correct.

- For the built-in Apple screen the Mac's model identifier (for example Mac14,9) is used and EveryMac's lookup page for that identifier opens; the list is shown as it is, because several configurations can share one identifier. An external monitor connected to a Mac is not treated as the built-in screen.
- For other monitors the model is copied and displayspecifications.com opens: that site's search cannot be filled from a link, so you paste the model into it.
- No serial number or personal identifier is sent; the application itself uses no internet here (your browser does), so without a connection the model simply stays on the clipboard.

A specification page is not a correction
The window explains four different things: panel technology (IPS, VA…), light-emitting technology (white LED, phosphor, quantum dot, OLED…), measurement mode, and correction (spectral or matrix). What a site says about a monitor helps to recognise it; it is not a validated correction for the combination of your monitor and your instrument, and nothing is selected for you.

11. Uninstall and where your data stays

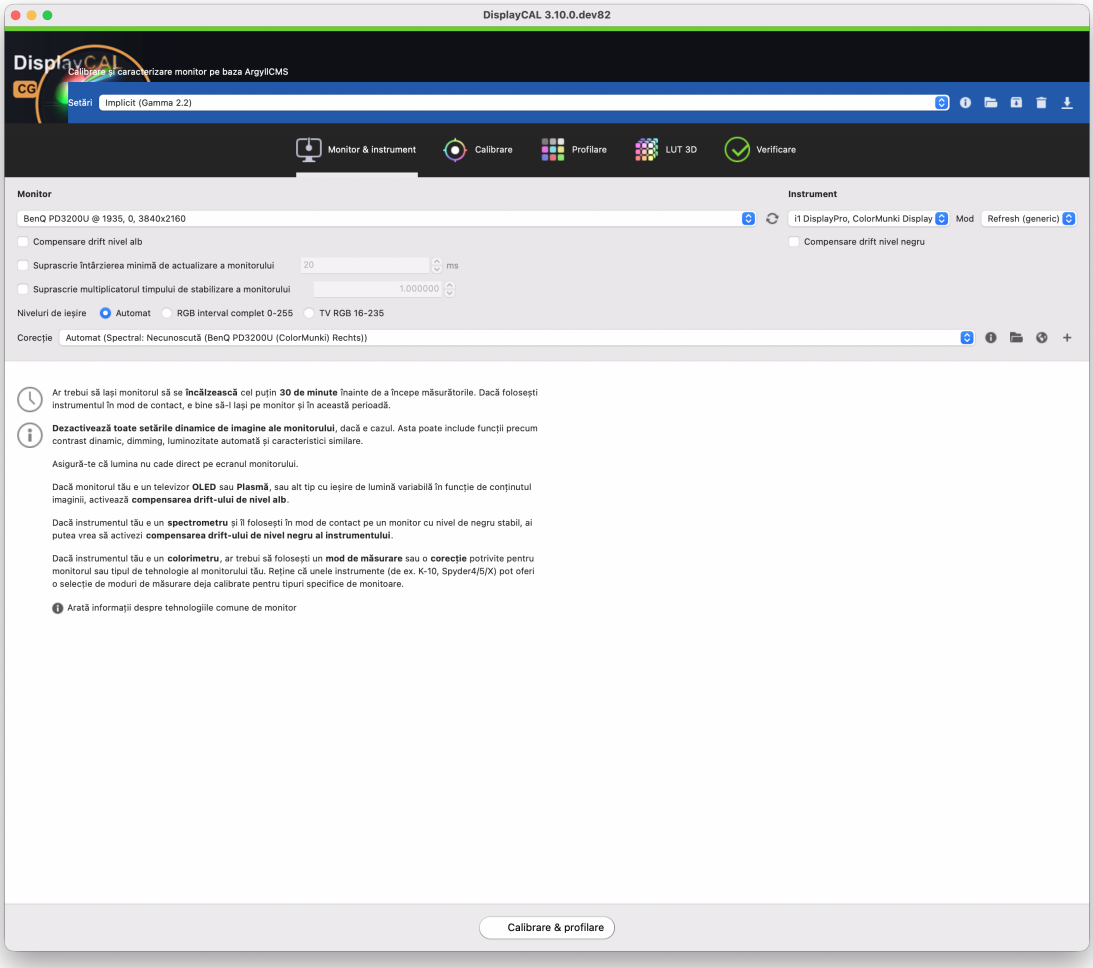
- macOS — quit the application, then move DisplayCAL and the other tools (Curve Viewer, ICC Profile Info, Testchart Editor, 3D LUT Maker, Scripting Client, Synthetic ICC, VRML to X3D) from Applications to the Bin.
- Windows — Settings → Apps → DisplayCAL-CG → Uninstall.

The application's data is not removed by either way, so you can reinstall without losing settings. To delete it too:

- macOS: `~/Library/Preferences/DisplayCAL` (settings), `~/Library/Application Support/DisplayCAL` (data and session records), `~/Library/Logs/DisplayCAL` (logs).
- Windows: `%APPDATA%\DisplayCAL` (settings, data and session records), `%LOCALAPPDATA%\DisplayCAL\Logs` (logs).

12. Display & instrument

The first tab — pick WHICH monitor you're calibrating and WITH WHICH instrument. The app needs to know both before you can move on to Calibration.



The Display & instrument tab, with an external monitor and an i1DisplayPro colorimeter already detected.

Field	What it does
Display	Pick the monitor to calibrate from the list if you have more than one connected. The round □ button next to it rescans connected displays.
Instrument	The colorimeter/spectrophotometer detected via USB. If it's empty, check the USB cable and that the instrument is recognized by the operating system.
Measurement mode	The instrument's measurement mode — "Refresh (generic)" suits most LCD/LED monitors. Some instruments (K-10, Spyder4/5/X) offer precalibrated modes for specific display types — pick the closest match if one exists.
White level drift compensation	Enable if the display is an OLED/Plasma TV or another type with variable light output depending on on-screen content.
Black level drift compensation	Enable if you're using a spectrometer in contact mode on a display with an unstable black level.

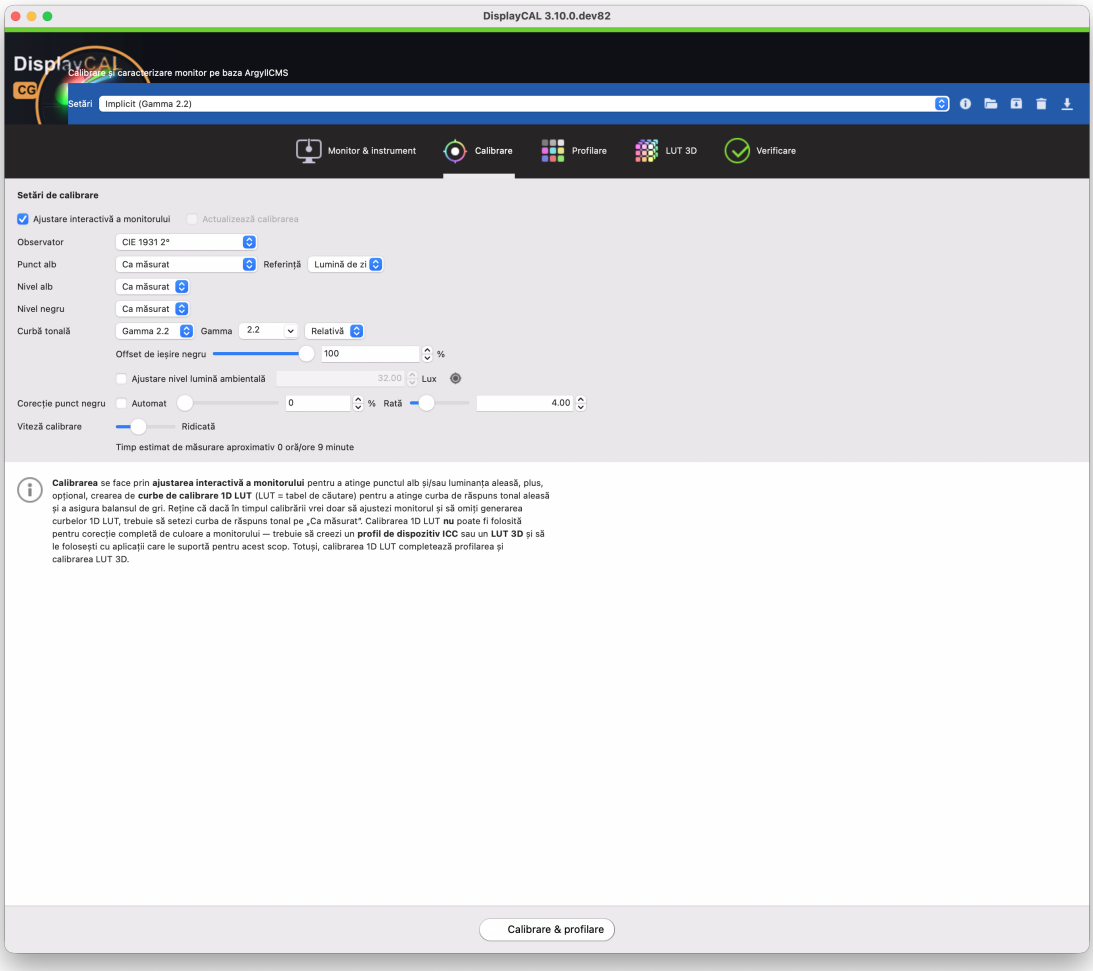
Field	What it does
Output levels	"Auto" is correct in nearly all cases. "TV RGB 16-235" only applies if the display/video card deliberately limits the signal range, as with a TV used as a monitor.
Correction	The instrument+display specific color correction — DisplayCAL-CG picks it automatically ("Auto (Spectral: ...)") when it can; don't change it manually unless you know exactly what you're doing.

Before you measure

Let the display warm up for at least 30 minutes before calibrating — a cold monitor's colors drift as it stabilizes thermally. Turn off any dynamic picture settings (dynamic contrast, auto brightness) and avoid light falling directly on the screen during measurement.

13. Calibration

The second tab — choose WHAT target state you want for the display: what white point, what luminance, what tone curve.



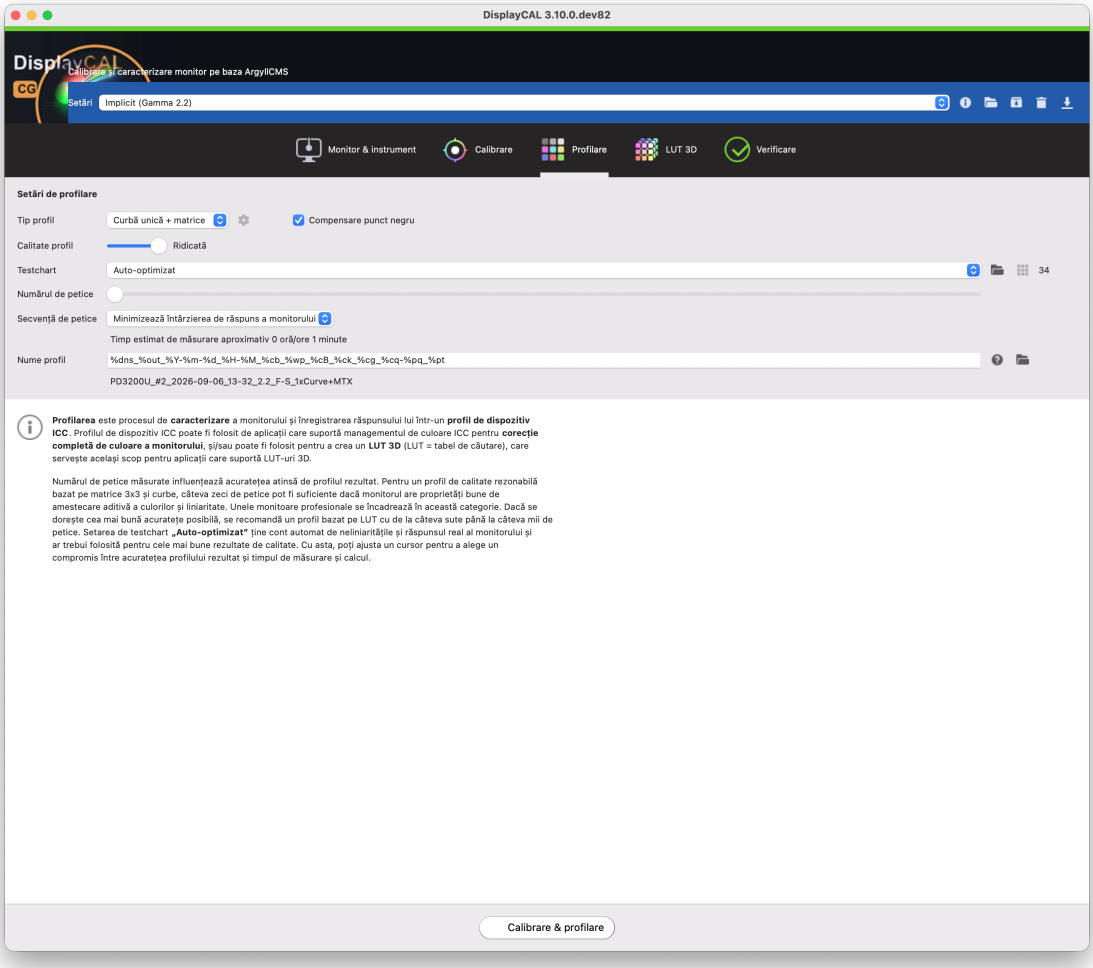
Default calibration settings (Gamma 2.2, white point and levels "as measured").

Field	What it does
Interactive display adjustment	When checked, the app guides you to manually adjust the monitor's physical controls (brightness, contrast, RGB) during calibration, to get as close as possible to the target before generating the software curves.
Observer	The CIE standard used to interpret color — "CIE 1931 2°" is the default, suitable for the vast majority of situations.
White point	"As measured" keeps the display's native white point. You can instead pick a fixed color temperature (e.g. 6500K/D65) if you need to hit an exact standard, with a reference ("Daylight"/"Black body").
White level / Black level	"As measured" keeps the display's native luminance. You can set a fixed value manually (e.g. 120 cd/m²) if you need to meet a luminance standard.
Tonal response curve	The shape the resulting tone response will have — "Gamma 2.2" is the default fit for photo/web; "Rec. 1886" or other curves matter mostly for video.
Black output offset	0% = "pure" black; 100% = black follows the chosen curve exactly with no offset. In-between values compensate for displays with elevated black levels.
Black point correction	The rate/percentage used to correct near-black nonlinearities — "Auto" lets the app decide.
Calibration speed	A time/accuracy trade-off — "High" (default) is enough for most uses.

1D LUT calibration doesn't replace an ICC profile
The curves generated here only correct the display's overall tonality — for full color correction you also need a device ICC profile or a 3D LUT, created in the following tabs.

14. Profiling

The third tab — DisplayCAL-CG displays actual color patches on screen, measures them with the instrument, and builds the ICC profile that characterizes your now-calibrated display.



Profiling settings — "Single curve + matrix" profile type, auto-optimized testchart, 34 patches.

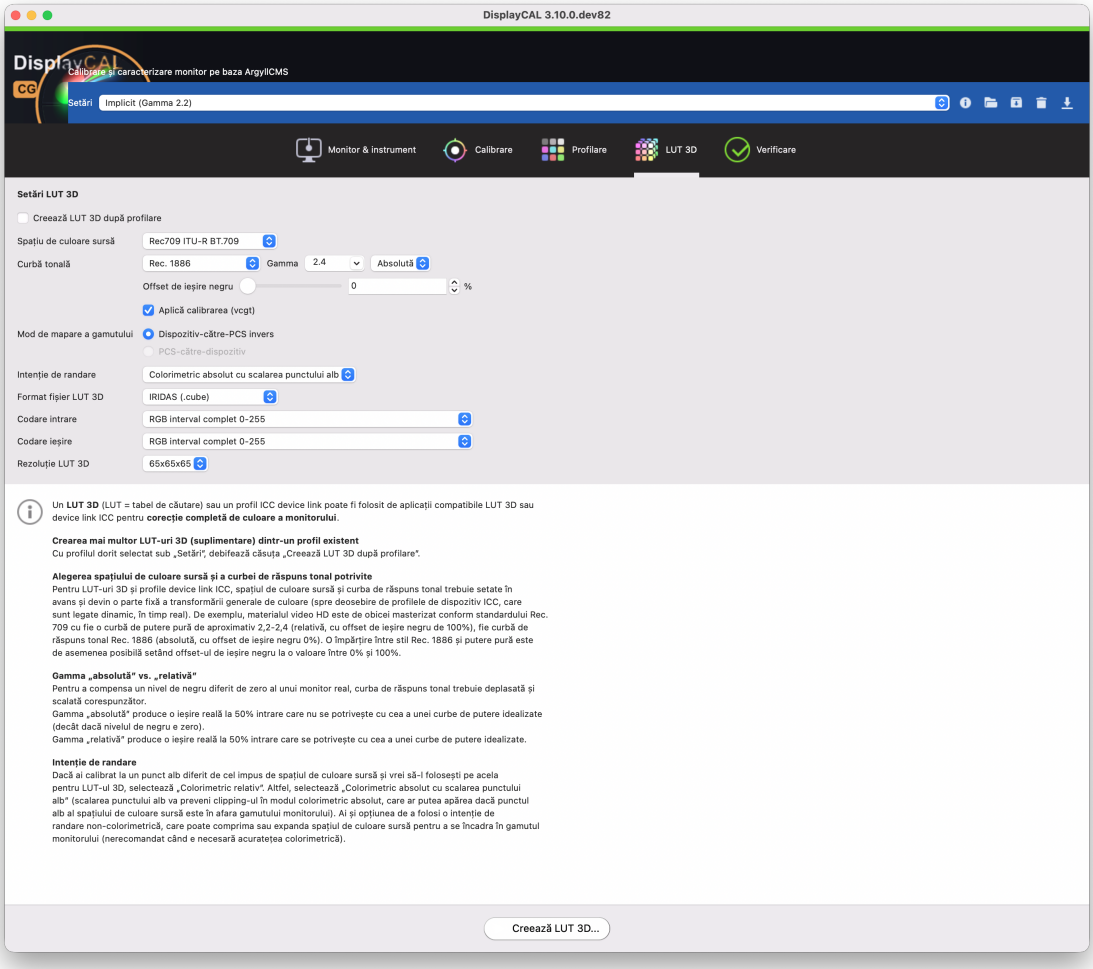
Field	What it does
Profile type	"Single curve + matrix" is fast and sufficient for many good displays; a LUT-based profile with hundreds to thousands of patches gives the best possible accuracy, but takes much longer.
Black point compensation	Recommended checked — improves accuracy in dark areas.
Profile quality	Low→High slider — affects how finely the profile is computed from the measured data, not the number of patches.
Testchart	"Auto-optimized" automatically picks patch distribution accounting for your display's actual nonlinearities — recommended for the best results.
Number of patches	The slider controls how many patches get measured — more patches = a more accurate profile, but a longer measurement.
Patch order	"Minimize display response delay" orders the patches to shorten total measurement time.

Field	What it does
Profile name	The automatic naming template — you can freely edit the field below if you want a custom profile name.

> **How long it takes**
> The app shows an estimated time under the profiling settings (e.g. "about 1 minute" for 34 patches) — a LUT-based profile with thousands of patches can take anywhere from a few minutes to over an hour.

15. 3D LUT

An optional tab — generates a 3D LUT (a three-dimensional lookup table) from the profile already created, for applications that support color correction via 3D LUT instead of an ICC profile (common in video/color-grading workflows — DaVinci Resolve, media players).



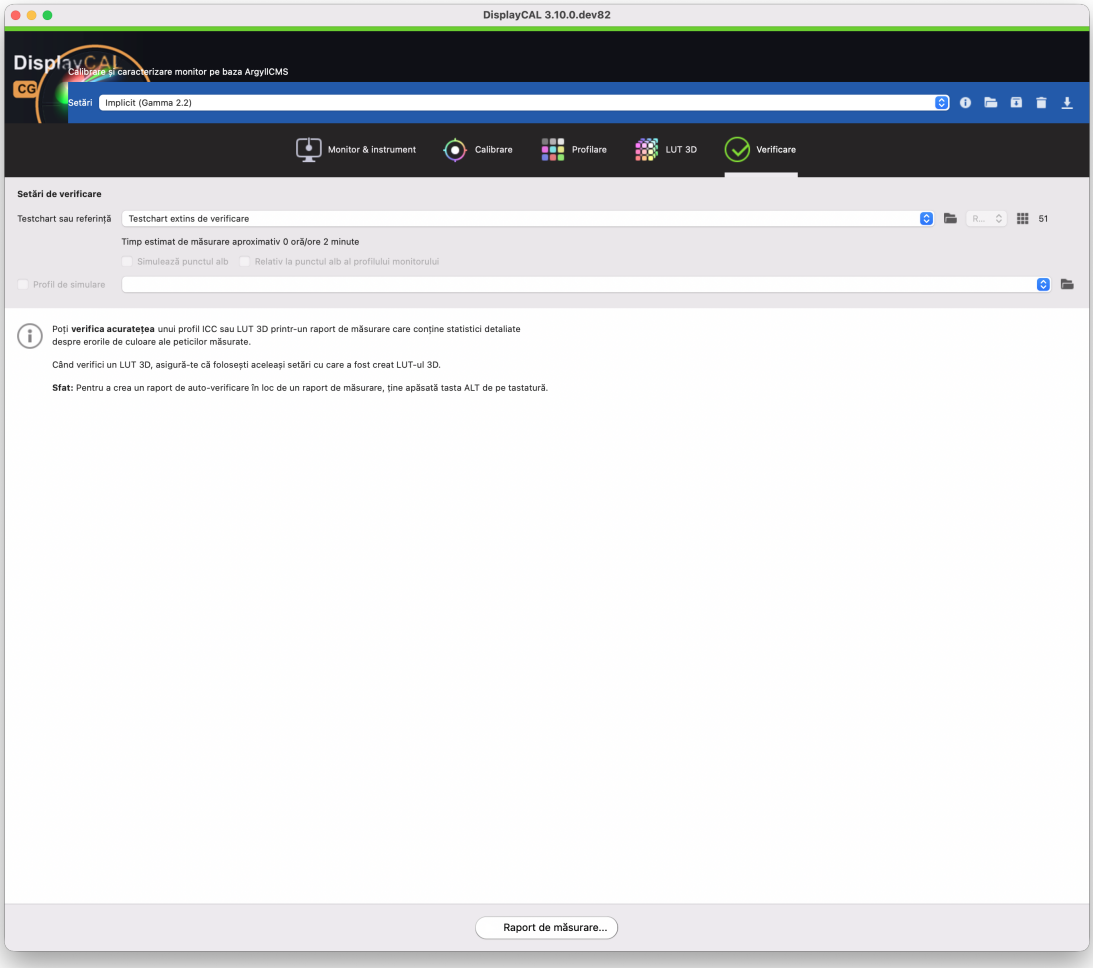
3D LUT settings — Rec709 source, Rec.1886 curve, IRIDAS .cube format, 65×65×65 resolution.

Field	What it does
Create 3D LUT after profiling	Check this if you want the LUT to be generated automatically right after profiling finishes, as one continuous step.
Source colorspace	The colorspace of the material you'll be playing back (e.g. "Rec709 ITU-R BT.709" for standard HD video).
Tone curve	Must match the source material's standard — HD video typically uses either a ~2.2-2.4 power curve or "Rec. 1886".
Gamut mapping mode	"Inverse device-to-PCS" is the standard choice for a display LUT (not a content-conversion LUT).
Rendering intent	"Absolute colorimetric with white point scaling" is recommended if you haven't explicitly calibrated to the source material's white point.
3D LUT file format	IRIDAS .cube is the most widely compatible (Resolve, most media players); other formats exist for specific software.
3D LUT resolution	65×65×65 is a good accuracy/file-size trade-off — higher resolutions increase both precision and file size.

Use the SAME settings the LUT was created with
When later verifying an already-created 3D LUT (Verification tab), make sure you use exactly the same settings (source space, curve, rendering intent) — otherwise the verification result is meaningless.

16. Verification

The fifth tab — checks how accurate an already created ICC profile or 3D LUT is, via a measurement report with statistics on the color errors measured across a set of patches.



Verification settings — extended verification testchart, 51 patches, ~2 minutes estimated.

Field	What it does
Testchart or reference	The patch set used for verification — "Extended verification testchart" is a standard set, independent of the one used for profiling (otherwise the verification would be biased).
Simulate white point	Compares the result relative to a simulated white point, instead of the display's native one.
Relative to display profile white point	Similar to the above, but relative to the white point RECORDED in the current profile.
Simulation profile	Optional — checks how the display would behave if it simulated another profile/colorspace.

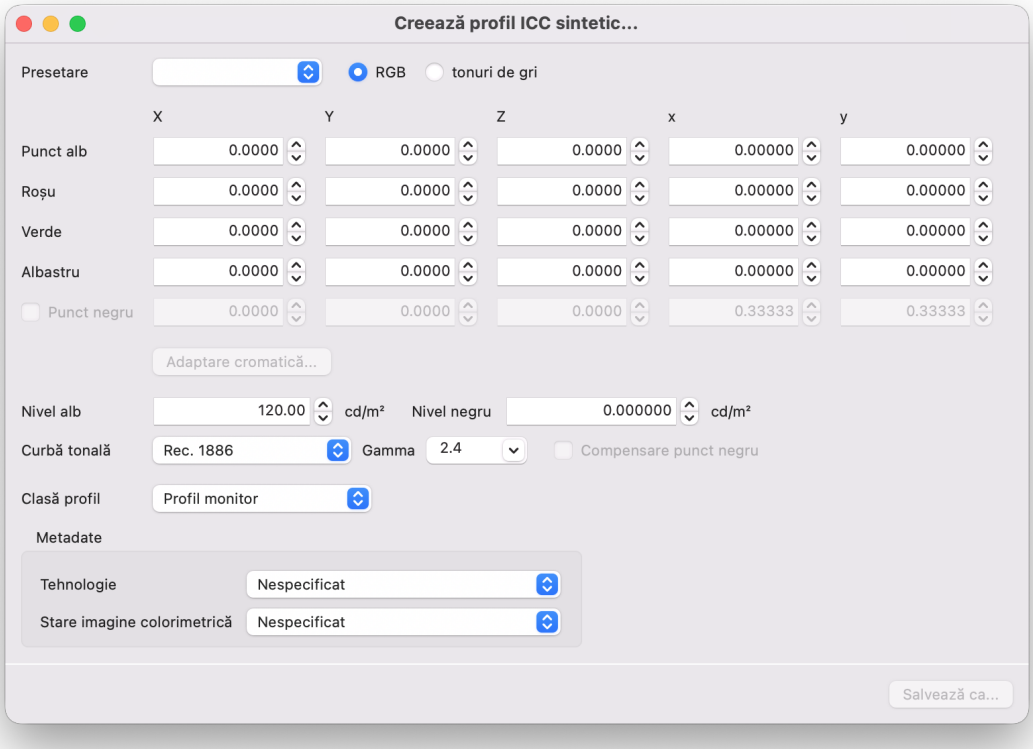
1. Pick the verification testchart (the default suits almost every case).
2. Click Measurement report... at the bottom of the window.
3. Follow the instrument on screen as the app displays and measures the test patches one by one.
4. At the end, a report opens with the average/maximum color errors measured (ΔE) — the lower the ΔE, the more accurate the profile.

> **Tip**
> Hold down the ALT key on your keyboard when clicking "Measurement report..." to create a self-check report instead of a regular measurement report.

17. Advanced tools

Besides the main workflow (the 5 tabs above), DisplayCAL-CG includes several standalone tools, useful for special cases — accessible from the main app's menu or as separate applications installed alongside it.

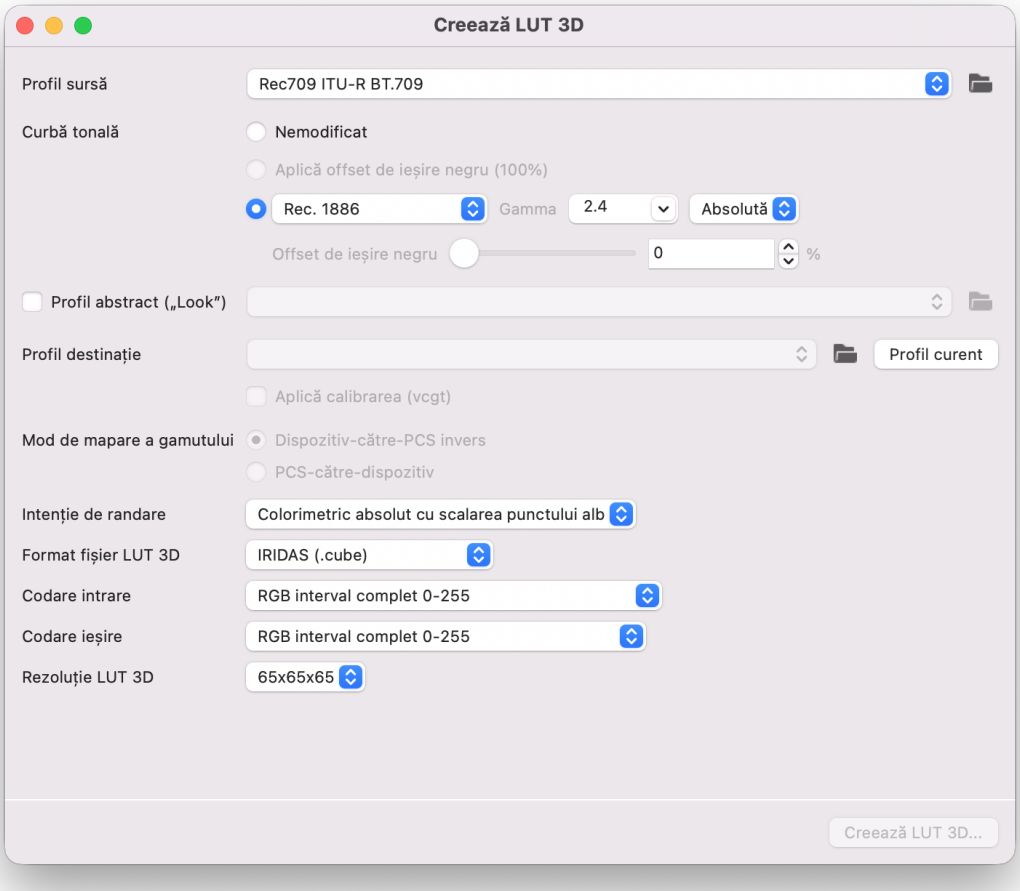
Create synthetic ICC profile



The synthetic ICC profile creation tool, built from manually described parameters (not from real measurements).

Builds an ICC profile from manually described parameters (white point, gamma, color primaries) — useful for generating a theoretical reference profile without measuring a real display (e.g. for simulation or testing).

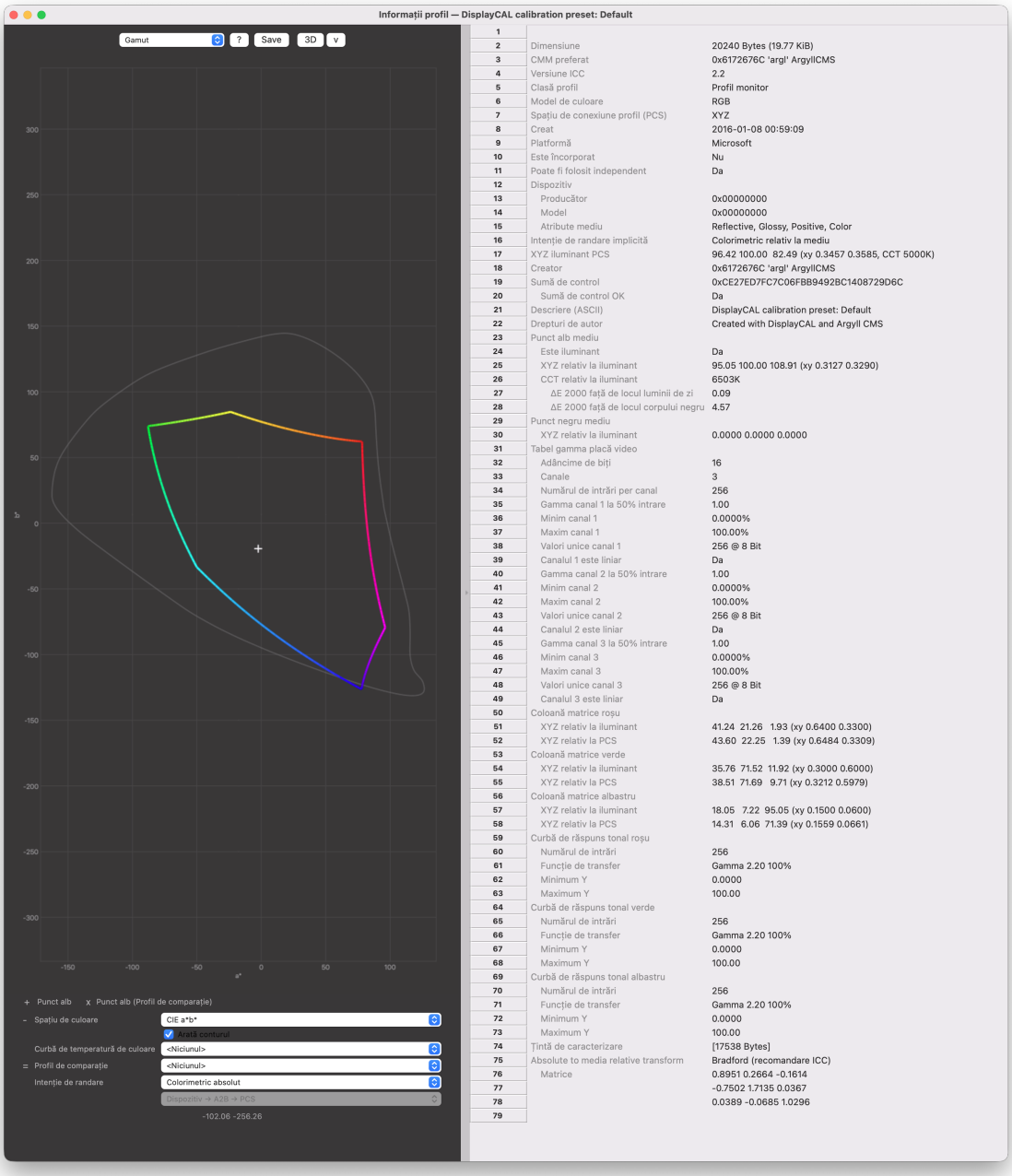
Create 3D LUT (standalone)



The standalone 3D LUT creation tool, for converting between colorspaces without going through a full display-calibration workflow.

The same 3D LUT generation logic as the "3D LUT" tab, but run independently of any display calibration workflow — useful for converting between two arbitrary profiles/colorspaces.

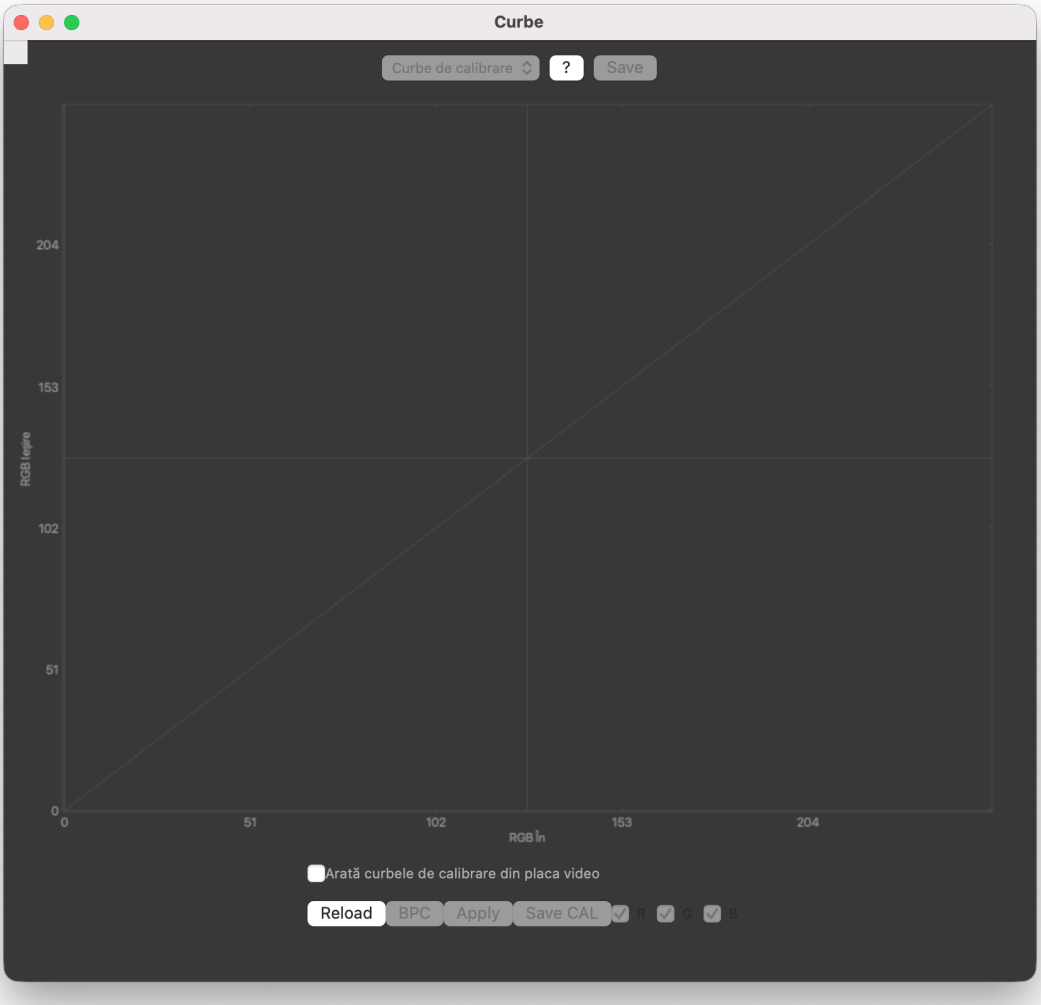
Profile Info



The Profile Info window — complete information about an ICC profile, plus a graphical representation of its gamut.

Opens any ICC profile (created by DisplayCAL-CG or from another source) and shows all the information it contains — white point, tone curve, color primaries — plus a 3D graphical representation of the covered color gamut.

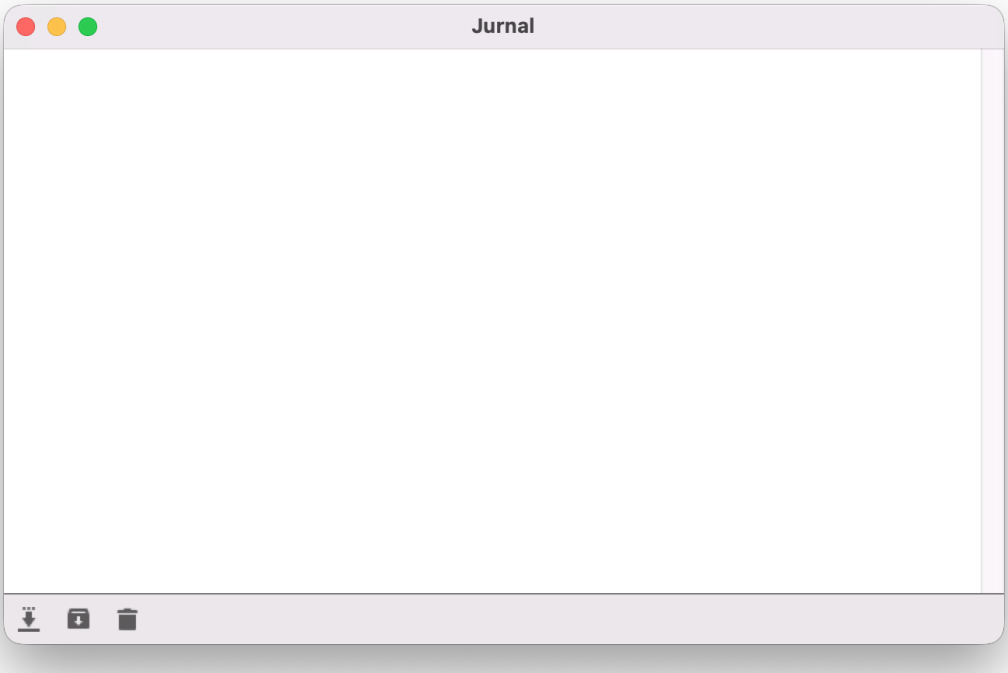
Curves



The Curves window — a visualization of the calibration curves (vcgt) currently loaded into the video card.

Graphically shows the calibration curves (VCGT — Video Card Gamma Table) currently loaded into the video card — useful for a quick visual check of what calibration is active right now.

Log



The Log window — the detailed technical log of the ArgyllCMS operations behind the application.

The detailed technical log of the ArgyllCMS commands run behind the scenes by the app — mainly useful for troubleshooting if something isn't working as expected and you want to understand exactly what happened.

18. GPLv3 license & optional support

> **100% free, forever**
> DisplayCAL-CG is free software, licensed under GPLv3 — fully functional from day one, no activation, no time-limited trial, no feature locked behind a payment. You can install, use and redistribute the application freely, subject to the included GPLv3 license terms (LICENSE.txt).

DisplayCAL-CG is built on the open-source work of DisplayCAL (Florian Höch) and its community continuators — full credits remain visible in the app (Help → About).

If the app has been useful to you, an optional support message appears occasionally in the application — it's purely informational, never a requirement to use any feature.

19. Support, feedback and credits

At the end of the flow, and in Help and About, two voluntary actions are offered: giving the project a star on GitHub (it only opens the repository page in your browser) and sending feedback (a small form that opens a pre-filled GitHub issue in your browser; you see what is included first, and the optional diagnostic has a preview and has personal paths removed). The application sends nothing itself and asks for no account. The invitation can be closed, or turned off for good.

Credits: DisplayCAL by Florian Höch, the displaycal-py3 continuation and its contributors, the ArgyllCMS engine by Graeme Gill, and this edition by Cristi Gordaș. Licence: GPLv3.

20. Updates

DisplayCAL-CG automatically checks at startup whether a newer version is available on the download page. You can also check manually from the Help menu.

1. If a new-version notification appears, click the link in it — it takes you straight to the download page with the latest version.
2. Download the new package (.pkg on Mac, .exe on Windows).
3. Install it over the current version, exactly as with the first install (see the "Installation" chapter) — your settings and already-created profiles remain untouched.
4. Restart the application after installing.